

SSH Assignment

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Case 1: Mac and Ubuntu on the same network



Step 1: Enable SSH on Ubuntu

On Ubuntu, install and enable SSH:

```
sudo apt update
sudo apt install openssh-server
sudo systemctl enable ssh
sudo systemctl start ssh
```

Check it's running:

```
sudo systemctl status ssh
```



Step 2: Find Ubuntu's local IP address

Run:

```
hostname -I
```

```
ashishkumar@ashishkumar:~$ hostname -I
< .0.2.15 fd17:625c:f037:2:c071:fc17:5e62:8435 fd17:625c:f037:2:a00:27ff:fe97:1cd5 >
ashishkumar@ashishkumar:~$
```



Step 3: Connect from Mac

On your Mac, open Terminal and run:

```
ssh username@192.168.1.42
```

Replace username with your Ubuntu username.

Enter your password when prompted — you're in! 

```
ashishkumar@ashishkumar:~$ ssh reyan08@10.0.2.15
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:G1ybnhtyBVHVw/mDIoJ7WZq5VEN/sgIZYo+kxxHtFp8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.
reyan08@10.0.2.15's password:
Permission denied, please try again.
reyan08@10.0.2.15's password:
Permission denied, please try again.
reyan08@10.0.2.15's password:
reyan08@10.0.2.15: Permission denied (publickey,password).
ashishkumar@ashishkumar:~$
```

ON MAC

```
Code9.c
College-Authorization-letter-SIH2025.docx
combined.txt
C_Patterns
Desktop
Documents
Downloads
ds_starter
'Engineering Physics by Hitendra K Malik A K Singh (z-lib.org).pdf'
f1.txt
factorial.sh
fdkjin
file1.txt
file2.txt
folder
g2.png
HelloWorld.class
j1.java
JoplinBackup
LAB0.pdf
lab1.md
LAB1.pdf
lab2.md
LAB2.pdf
lab3.md
LAB3.pdf
'lab4 (1).md'
lab4.md
LAB4.pdf
LAB5.pdf
Lab6_Shell_Scripting_Basics.md
Linux_lab
Linux Lab
qw13.sh
qw13.shx
qw15.sh
qw17.sh
qw2.sh
qw3.sh
qw4.sh
qw5.sh
qw6.sh
qw7.sh
qw8.sh
qw9.sh
school.pdf
'Screenshot From 2025-09-24 16-07-03.png'
scripts
_sh.1
Shell
snake.c
snap
starterkit
starter_kit.sh
starter_kit_streamlit.sh
temp
Templates
testfile1.txt
testfile.txt
transfertest.txt
Videos
w.sh.save
ws.txt
zf.py
zxn1.sh
zxn2.sh
```

Case 2: Mac and Ubuntu on different networks (e.g., home ↔ office or remote server)

♦ Option 1: Connect to a remote Ubuntu server (e.g., cloud VM)

If your Ubuntu machine is a remote server (AWS, DigitalOcean, etc.), it already has SSH enabled. You just need:

```
ssh username@<public-IP-address>
```

Make sure:

The server's firewall allows port 22 (default SSH port).

Your cloud provider security group or firewall allows inbound SSH.

✔ ♦ Option 2: Connect to your Ubuntu home PC remotely

If your Ubuntu computer is behind a router:

1. Find your Ubuntu's local IP:

`hostname -I`

```
ashishkumar@ashishkumar:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:c071:fc17:5e62:8435 fd17:625c:f037:2:a00:27ff:fe97:1cd5
ashishkumar@ashishkumar:~$
```

2. Log in to your router and port-forward port 22 to that IP.
3. Find your public IP (visit <https://whatismyipaddress.com>)
4. From your Mac, connect via:

```
ssh username@<your-public-IP>
```

or better yet, use a dynamic DNS (DDNS) service like No-IP or DuckDNS so you can use a hostname instead of changing IPs.

```
ashishkumar@ashishkumar:~$ sudo apt install curl
[sudo] password for ashishkumar:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 84 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.6 [226 kB]
Fetched 226 kB in 3s (89.3 kB/s)
Selecting previously unselected package curl.
(Reading database ... 204963 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
ashishkumar@ashishkumar:~$
```

```

ashishkumar@ashishkumar:~$ sudo systemctl start ssh
ashishkumar@ashishkumar:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-03 18:25:45 IST; 4h 59min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 9512 (sshd)
      Tasks: 1 (limit: 6845)
     Memory: 1.3M (peak: 19.8M)
        CPU: 441ms
    CGroup: /system.slice/ssh.service
            └─9512 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 03 19:12:15 ashishkumar sshd[11942]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15 user=reyan08
Nov 03 19:12:18 ashishkumar sshd[11942]: Failed password for reyan08 from 10.0.2.15 port 35672 ssh2
Nov 03 19:12:32 ashishkumar sshd[11942]: Failed password for reyan08 from 10.0.2.15 port 35672 ssh2
Nov 03 19:12:47 ashishkumar sshd[11942]: Failed password for reyan08 from 10.0.2.15 port 35672 ssh2
Nov 03 19:12:48 ashishkumar sshd[11942]: Connection closed by authenticating user reyan08 10.0.2.15 port 35672 [preauth]
Nov 03 19:12:48 ashishkumar sshd[11942]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15 user=reyan08
Nov 03 19:16:21 ashishkumar sshd[12123]: Accepted password for reyan08 from 10.0.2.15 port 53940 ssh2
Nov 03 19:16:21 ashishkumar sshd[12123]: pam_unix(sshd:session): session opened for user reyan08(uid=1002) by reyan08(uid=0)
Nov 03 19:21:55 ashishkumar sshd[12713]: Accepted password for reyan08 from 10.0.2.15 port 45664 ssh2
Nov 03 19:21:55 ashishkumar sshd[12713]: pam_unix(sshd:session): session opened for user reyan08(uid=1002) by reyan08(uid=0)
lines 1-23/23 (END)...skipping...
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-03 18:25:45 IST; 4h 59min ago
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     Docs: man:sshd(8)
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    CGroup: /system.slice/ssh.service
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```

```

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● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-03 18:25:45 IST; 4h 59min ago
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Nov 03 19:12:48 ashishkumar sshd[11942]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.15 user=reyan08
Nov 03 19:16:21 ashishkumar sshd[12123]: Accepted password for reyan08 from 10.0.2.15 port 53940 ssh2
Nov 03 19:16:21 ashishkumar sshd[12123]: pam_unix(sshd:session): session opened for user reyan08(uid=1002) by reyan08(uid=0)
Nov 03 19:21:55 ashishkumar sshd[12713]: Accepted password for reyan08 from 10.0.2.15 port 45664 ssh2
Nov 03 19:21:55 ashishkumar sshd[12713]: pam_unix(sshd:session): session opened for user reyan08(uid=1002) by reyan08(uid=0)

```



GUI

✓ 1) Prepare & secure the remote machine (run on friend's laptop)

Run these on the friend's laptop (they must have sudo):

```

# install SSH and optional VNC
sudo apt update
sudo apt install -y openssh-server tigervnc-standalone-server

# create non-root user if needed (won't fail if exists)
sudo useradd -m frienduser || true

```

```
# enable and start SSH
sudo systemctl enable --now ssh

# add your public key (replace <your-pubkey>) and set safe permissions
sudo -u frienduser mkdir -p /home/frienduser/.ssh
echo "<your-pubkey>" | sudo tee -a /home/frienduser/.ssh/authorized_keys
sudo chmod 700 /home/frienduser/.ssh
sudo chmod 600 /home/frienduser/.ssh/authorized_keys
sudo chown -R frienduser:frienduser /home/frienduser/.ssh
```

Verify sshd is running:

```
sudo systemctl status ssh --no-pager
# or
ss -tlnp | grep :22
```

✅ 2) Test X11 forwarding (single GUI app) — from your laptop

Preflight on your laptop:

Linux: usually ready.

macOS: install & run XQuartz and start it before SSH.

Windows: install an X server like VcXsrv or Xming and run it.

Then on your laptop run:

```
# connect with X11 forwarding (use -X or -Y if needed; -Y is more
permissive)
ssh -p 22 -X frienduser@FRIEND_IP
# once connected, run a simple GUI test:
xeyes &    # or gedit & or xclock &
```

Success criteria: the GUI app window appears on your laptop and is responsive.

Troubleshooting quick checks:

If you see Warning: No xauth data; using fake or X apps fail: try -Y (trusted) instead of -X: `ssh -Y -p 22 frienduser@FRIEND_IP`.

Ensure DISPLAY is set on the SSH session (run `echo $DISPLAY` — should show something like `localhost:10.0`).

Ensure client X server (XQuartz/VcXsrv) is running and allowing connections.

Security note: X11 forwards every GUI app over SSH, but X11 is older and can let remote apps access local X resources — use only with trusted machines. For stronger isolation, prefer the VNC-over-SSH approach below.

✓ 3) Full desktop: VNC server on friend's laptop + SSH tunnel from your laptop

On friend's laptop (as frienduser):

```
# start a VNC server on display :1 (creates :1 -> TCP 5901)
vncserver :1
# optionally stop with: vncserver -kill :1
```

(Configure desktop environment in ~/.vnc/xstartup if needed — many distros auto-configure.)

On your laptop: create a local SSH tunnel (keeps VNC server bound to localhost on remote; only SSH port open externally)

```
# create an SSH tunnel mapping your localhost:5901 to friend's
localhost:5901
ssh -L 5901:localhost:5901 -p 22 frienduser@FRIEND_IP -N &
# -N = no remote command; & runs in background (adjust as preferred)
```

Then open your VNC viewer and connect to:

```
localhost:5901
```